

Shape-Scan Phase 2: Chrome Subsystem + Liquid Glass Retrofit

| Session [abdbe30a](#) — Continuation of [ec52e00b](#)

What Was Built

Step 7B — Chrome Subsystem

Completed the observability layer by populating all previously-empty sparkline and mini-viz canvases in the Component Breakdown Dashboard.

Card	Sparkline	Mini-Viz	Technique
TFEA	Line chart (<i>pre-existing</i>)	Heatmap strip (<i>pre-existing</i>)	—
Markov	3-bar: $H_2 / H / \bar{H}_r$	3-bar: density / pairs / $\sigma(H)$	<code>drawMultiBar()</code>
TCGE	3-bar: $B/E / \rho / SCC$	4-bar: $\bar{d} / \max / CC / \mathfrak{C}$	<code>drawMultiBar()</code>
AISE	10-spoke radar chart	Radial gauge (<i>pre-existing</i>)	<code>drawIntentRadar()</code>
CQSF	—	Radial gauge (<i>pre-existing</i>)	—

HUD Viewport Badges: Three glass pills overlaying the primary viewport (bottom-right):

- **Entropy** (cyan) — mean entropy value
- **Threat** (red) — composite threat percentage
- **Verdict** (color-coded) — quarantine verdict with icon

Step 7 — Liquid Glass Retrofit

Full DS-001-G-D-LGT design system audit and retrofit, informed by 4 web research passes on 2026 CSS glassmorphism best practices.

Research Synthesis

- iOS 26 Liquid Glass: asymmetric inset shadows, `saturate(180%)`, ambient orbs behind glass
- Dark mode glass: tinted-dark backgrounds (NOT flat black), cool-toned borders
- Performance: `isolation: isolate` instead of `contain: paint` (avoid backdrop root)
- Fallbacks: `@supports not (backdrop-filter)` for accessibility

Changes Applied (Layered)

Layer 1 — Foundation

- New design tokens: `--radius-glass: 20px`, `--radius-pill: 999px`, full glass material vars
- Body background: multi-stop gradient (`145deg deep-space`) replaces flat `#0a0e17`
- 3 ambient orbs (2× pseudo-elements + 1× DOM div) with 26s/32s/38s drift cycles

Layer 2 — Glass Material

- `.glass-panel`: Full asymmetric inset shadow treatment + `blur(24px) saturate(180%)` + 20px radius
- `@supports` fallback for non-glass environments
- `isolation: isolate` for stacking context (NOT `contain: paint`)

Layer 3 — Chrome Elements

- Header: Sticky glass strip with `blur(20px)` + inset shadows
- Scan button: Pill radius + cyan glow shadow
- Browse button: Full glass material (blur + inset shadows)
- Input field: Inset shadow + darker glass bg

Layer 4 — Component Polish

- Flag badges: Pill radius + glass backdrop
- Viewport controls: Glass icon buttons
- Viewport slots: Glass border + inset shadow + 12px radius
- Viewport labels: Glass pill badges with backdrop-filter
- Component card hover: DS-001 asymmetric shadow + cyan tint
- Verdict icon: Glass circle with inset shadow
- Detail lens close: Round glass button
- Graph container: Glass border + consistent radius
- Fingerprint section: Subtle separator

Proactive Fixes Applied

1. **Backdrop root prevention:** Replaced `contain: layout style paint` with `isolation: isolate` — research showed `contain: paint` creates a "backdrop root" that limits what `backdrop-filter` can blur
2. **HUD badge positioning:** Moved to bottom-right to avoid collision with viewport-label at bottom-left
3. **Z-index hierarchy:** Verified stacking order (orbs:0 → main:1 → badges:6 → header:50)
4. **Mobile performance:** Reduced orb opacity to 0.5 on screens ≤768px

Remaining Steps

Step	Status	Description
2D — Focus Lens		2.5D wireframe detail view for byte-range inspection

All other Phase 2 steps are **COMPLETE**.

Files Modified

Frontend

- [main.css](#) — Full Liquid Glass retrofit (25 individual changes)
- [component_dashboard.js](#) — Chrome Subsystem renderers + HUD population
- [index.html](#) — HUD badge elements + ambient orb div

No Rust Changes

All modifications were frontend-only (CSS/HTML/JS). Backend compiled in 1.33s (incremental).

Validation

- Build:  Compiles without errors (1 warning: unused `scan_count`)
- Stability:  No restart loops, app runs persistently
- Performance:  `isolation: isolate` used instead of `contain: paint` to preserve backdrop-filter
- Responsive:  Glass radius reduces on mobile, orbs dim for performance